

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

A chemist studying the impact of salt on a process mixes x kilograms of a low-salt mixture, which is **2%** salt by weight, with y kilograms of a high-salt mixture, which is **96%** salt by weight, to create **24** kilograms of a mixture that is **4%** salt by weight. Which equation represents this situation?

Answer

- A. $0.96x + 0.02y = (0.04)(24)$
- B. $0.02x + 0.96y = (0.04)(24)$
- C. $0.96x + 0.02y = 24$
- D. $0.02x + 0.96y = 24$

Correct Answer: B

Rationale

Choice B is correct. It's given that a chemist mixes x kilograms of a low-salt mixture, which is **2%** salt by weight. Multiplying **0.02** by the amount of the low-salt mixture, x kilograms, yields **$0.02x$** kilograms of salt in the low-salt mixture. It's also given that the chemist mixes y kilograms of a high-salt mixture, which is **96%** salt by weight. Multiplying **0.96** by the amount of the high-salt mixture, y kilograms, yields **$0.96y$** kilograms of salt in the high-salt mixture. Therefore, the total amount of salt in the combined mixture is **$0.02x + 0.96y$** kilograms. It's given that the low-salt mixture and the high-salt mixture together create **24** kilograms of a combined mixture that is **4%** salt by weight. Thus, the amount of salt in the combined mixture is **$0.04(24)$** kilograms. Since the total amount of salt in the combined mixture equals the amount of salt in the low-salt mixture and the amount of salt in the high-salt mixture, the equation **$0.02x + 0.96y = (0.04)(24)$** represents this situation.

Choice A is incorrect. This equation represents a situation where the low-salt mixture is **96%**, not **2%**, salt by weight and the high-salt mixture is **2%**, not **96%**, salt by weight.

Choice C is incorrect. This equation represents a situation where the low-salt mixture is **96%**, not **2%**, salt by weight and the high-salt mixture is **2%**, not **96%**, salt by weight, and where the combined mixture contains **24** kilograms of salt, not **24** kilograms of a mixture that is **4%** salt by weight.

Choice D is incorrect. This equation represents a situation where the combined mixture contains **24** kilograms of salt, not **24** kilograms of a mixture that is **4%** salt by weight.